

Full/Low Speed 2.5 kV USB Digital Isolator

FEATURES

- ▶ USB 2.0 compatible
- ▶ Low and full speed data rate: 1.5 Mbps and 12 Mbps
- ▶ Bidirectional communication
- ▶ 4.5 V to 5.5 V V_{BUS} operation
 - ▶ 7 mA maximum upstream supply current at 1.5 Mbps
 - ▶ 8 mA maximum upstream supply current at 12 Mbps
 - ▶ 2.3 mA maximum upstream idle current
- ▶ Upstream short-circuit protection
- ▶ Class 3A contact ESD performance per ANSI/ESD STM5.1-2007
- ▶ High temperature operation: 105°C
- ▶ High common-mode transient immunity: >25 kV/μs
- ▶ 16-lead SOIC wide body package
- ▶ RoHS compliant
- ▶ AEC-Q100 qualified for automotive applications
- ▶ Safety and regulatory approvals
 - ▶ UL 1577
 - ▶ $V_{ISO} = 2500 V_{RMS}$ for 1 minute
 - ▶ IEC/EN/CSA 62368-1
 - ▶ IEC/CSA 61010-1
 - ▶ CQC GB4943.1
 - ▶ DIN EN IEC 60747-17 (VDE 0884-17)
 - ▶ $V_{IORM} = 645 V_{PEAK}$

APPLICATIONS

- ▶ USB peripheral isolation
- ▶ Isolated USB hub

FUNCTIONAL BLOCK DIAGRAM

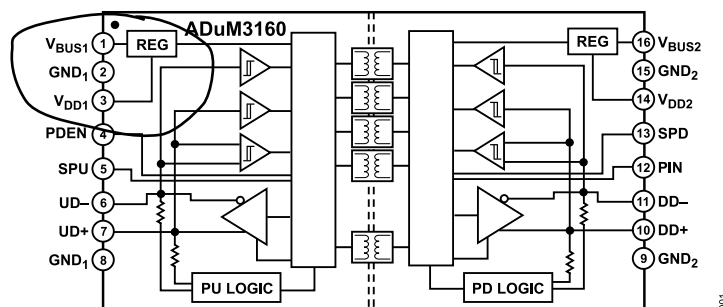


Figure 1.

GENERAL DESCRIPTION

The ADuM3160¹ is a USB port isolator based on Analog Devices, Inc., iCoupler® technology. Combining high speed CMOS and monolithic air core transformer technology, this isolation component provides outstanding performance characteristics and is easily integrated with low and full speed USB-compatible peripheral devices.

Many microcontrollers implement USB so that it presents only the D+ and D- lines to external pins. This is desirable in many cases because it minimizes external components and simplifies the design; however, this presents particular challenges when isolation is required. Because the USB lines must switch between actively driving D+/D- and allowing external resistors to set the state of the bus, the ADuM3160 provides mechanisms for detecting the direction of data flow and control over the state of the output buffers. Data direction is determined on a packet-by-packet basis.

The ADuM3160 uses the edge detection based iCoupler technology in conjunction with internal logic to implement a transparent, easily configured, upstream-facing port isolator. Isolating the upstream port provides several advantages in simplicity, power management, and robust operation.

The isolator has a propagation delay comparable to that of a standard hub and cable. It operates with the bus voltage on either side ranging from 4.5 V to 5.5 V, allowing connection directly to V_{BUSx} by internally regulating the voltage to the signaling level. The ADuM3160 provides isolated control of the pull-up resistor to allow the peripheral to control connection timing. The device draws low enough idle current that a suspend state is not required. A 5 kV, reinforced insulation version, the ADuM4160, is available.



¹ Protected by U.S. Patents 5,952,849; 6,873,065; 7,075,329; and 8,432,812.

TABLE OF CONTENTS

Features.....	1	Functional Description.....	11
Applications.....	1	Product Usage.....	11
General Description.....	1	Compatibility of Upstream Applications.....	12
Functional Block Diagram.....	1	Power Supply Options.....	12
Specifications.....	3	Printed Circuit Board Layout.....	12
Electrical Characteristics.....	3	DC Correctness and Magnetic Field	
Package Characteristics.....	4	Immunity.....	12
Regulatory Information.....	5	Insulation Lifetime.....	13
Insulation and Safety-Related Specifications.....	5	Outline Dimensions.....	14
DIN EN IEC 60747-17 (VDE 0884-17)		Ordering Guide.....	14
Insulation Characteristics.....	6	Number of Inputs, Maximum Full Speed	
Recommended Operating Conditions.....	7	Data Rate, Maximum Full Speed	
Absolute Maximum Ratings.....	8	Propagation Delay, and Maximum Full	
ESD Caution.....	8	Speed Jitter Options	14
Maximum Continuous Working Voltage.....	8	Evaluation Boards.....	14
Pin Configuration and Function Descriptions.....	9	Automotive Products.....	15
Applications Information.....	11		

REVISION HISTORY**4/2025—Rev. C to Rev. D**

Changes to Features Section.....	1
Changes to Regulatory Information Section and Table 3.....	5
Changes to Table 4.....	5
Changed DIN V VDE V 0884-10 (VDE V 0884-10) Insulation Characteristics Section to DIN EN IEC 60747-17 (VDE 0884-17) Insulation Characteristics Section.....	6
Changes to DIN EN IEC 60747-17 (VDE 0884-17) Insulation Characteristics Section, Table 5, and Figure 2 Caption.....	6
Changes to Insulation Lifetime Section.....	13
Deleted Figure 8, Figure 9, and Figure 10; Renumbered Sequentially.....	13
Added Number of Inputs, Maximum Full Speed Data Rate, Maximum Full Speed Propagation Delay, and Maximum Full Speed Jitter Options.....	14

SPECIFICATIONS

ELECTRICAL CHARACTERISTICS

4.5 V ≤ V_{BUS1} ≤ 5.5 V, 4.5 V ≤ V_{BUS2} ≤ 5.5 V; 3.1 V ≤ V_{DD1} ≤ 3.6 V, 3.1 V ≤ V_{DD2} ≤ 3.6 V. All minimum/maximum specifications apply over the entire recommended operation range, unless otherwise noted; all typical specifications are at T_A = 25°C, V_{DD1} = V_{DD2} = 3.3 V. All voltages are relative to their respective grounds.

Table 1.

Parameter	Symbol	Min	Typ	Max	Unit	Test Conditions/Comments ¹
DC SPECIFICATIONS						
Total Supply Current ²						
1.5 Mbps						750 kHz logic signal rate, C _L = 450 pF
V _{DD1} or V _{BUS1} Supply Current	I _{DD1} (L)		5	7	mA	
V _{DD2} or V _{BUS2} Supply Current	I _{DD2} (L)		5	7	mA	
12 Mbps						6 MHz logic signal rate, C _L = 50 pF
V _{DD1} or V _{BUS1} Supply Current	I _{DD1} (F)		6	8	mA	
V _{DD2} or V _{BUS2} Supply Current	I _{DD2} (F)		6	8	mA	
Idle Current						
V _{DD1} or V _{BUS1} Idle Current	I _{DD1} (I)		1.7	2.3	mA	
Input Currents	I _{DD-} , I _{DD+} , I _{UD+} , I _{UD-} , I _{SPD} , I _{PIN} , I _{SPU} , I _{PDEN}	-1	+0.1	+1	μA	0 V ≤ V _{DD-} , V _{DD+} , V _{UD+} , V _{UD-} , V _{SPD} , V _{PIN} , V _{SPU} , V _{PDEN} ≤ 3.0 V
Single-Ended Logic High Input Threshold	V _{IH}	2.0			V	
Single-Ended Logic Low Input Threshold	V _{IL}			0.8	V	
Single-Ended Input Hysteresis	V _{HST}		0.4		V	
Differential Input Sensitivity	V _{DI}	0.2			V	V _{XD+} - V _{XD-}
Logic High Output Voltages	V _{OH}	2.8		3.6	V	R _L = 15 kΩ, V _L = 0 V
Logic Low Output Voltages	V _{OL}	0		0.3	V	R _L = 1.5 kΩ, V _L = 3.6 V
V _{DD1} and V _{DD2} Supply Undervoltage Lockout	V _{UVLO}	2.4		3.1	V	
V _{BUS1} Supply Undervoltage Lockout	V _{UVLOB1}	3.5		4.35	V	
V _{BUS2} Supply Undervoltage Lockout	V _{UVLOB2}	3.5		4.4	V	
Transceiver Capacitance	C _{IN}		10		pF	UD+, UD-, DD+, DD- to ground
Capacitance Matching			10		%	
Full Speed Driver Impedance	Z _{OUTH}	4		20	Ω	
Impedance Matching			10		%	
SWITCHING SPECIFICATIONS, I/O PINS, LOW SPEED						
Low Speed Data Rate			1.5		Mbps	C _L = 50 pF
Propagation Delay ³	t _{PHL} , t _{PLH}			325	ns	C _L = 50 pF, SPD = SPU = low, V _{DD1} , V _{DD2} = 3.3 V
Side 1 Output Rise/Fall Time (10% to 90%), Low Speed	t _{RL} , t _{FL}	75		300	ns	C _L = 450 pF, SPD = SPU = low, V _{DD1} , V _{DD2} = 3.3 V
Low Speed Differential Jitter, Next Transition	t _{LJN}		45		ns	C _L = 50 pF
Low Speed Differential Jitter, Paired Transition	t _{LJP}		15		ns	C _L = 50 pF
SWITCHING SPECIFICATIONS, I/O PINS, FULL SPEED						
Maximum Data Rate		12			Mbps	C _L = 50 pF
Propagation Delay ³	t _{PHL} , t _{PLH}	20	60	70	ns	C _L = 50 pF, SPD = SPU = high, V _{DD1} , V _{DD2} = 3.3 V
Output Rise/Fall Time (10% to 90%), Full Speed	t _{RF} , t _{FF}	4		20	ns	C _L = 50 pF, SPD = SPU = high, V _{DD1} , V _{DD2} = 3.3 V
Full Speed Differential Jitter, Next Transition	t _{HJN}		3		ns	C _L = 50 pF
Full Speed Differential Jitter, Paired Transition	t _{HJP}		1		ns	C _L = 50 pF
For All Operating Modes						

SPECIFICATIONS

Table 1. (Continued)

Parameter	Symbol	Min	Typ	Max	Unit	Test Conditions/Comments ¹
Common-Mode Transient Immunity at Logic High Output ⁴	$ CM_H $	25	35		kV/ μ s	$V_{UD+}, V_{UD-}, V_{DD+}, V_{DD-} = V_{DD1}$ or $V_{DD2}, V_{CM} = 1000$ V, transient magnitude = 800 V
Common-Mode Transient Immunity at Logic Low Output ⁴	$ CM_L $	25	35		kV/ μ s	$V_{UD+}, V_{UD-}, V_{DD+}, V_{DD-} = 0$ V, $V_{CM} = 1000$ V, transient magnitude = 800 V

¹ C_L = load capacitance, R_L = test load resistance, V_L = test load voltage, and V_{CM} = common-mode voltage.

² The supply current values are for the device running at a fixed continuous data rate 50% duty cycle, alternating J and K states. Supply current values are specified with USB-compliant load present.

³ Propagation delay of DD+ to UD+ or DD- to UD- in either signal direction is measured from the 50% level of the rising or falling edge to the 50% level of the rising or falling edge of the corresponding output signal.

⁴ CM_H is the maximum common-mode voltage slew rate that can be sustained while maintaining $V_O > 0.8 V_{DD2}$. CM_L is the maximum common-mode voltage slew rate that can be sustained while maintaining $V_O < 0.8$ V. The common-mode voltage slew rates apply to both rising and falling common-mode voltage edges. The transient magnitude is the range over which the common mode is slewed.

PACKAGE CHARACTERISTICS

Table 2.

Parameter	Symbol	Min	Typ	Max	Unit	Test Conditions/Comments
Resistance (Input to Output) ¹	R_{I-O}		10^{12}		Ω	
Capacitance (Input to Output) ¹	C_{I-O}		2.2		pF	$f = 1$ MHz
Input Capacitance ²	C_I		4.0		pF	
IC Junction-to-Ambient Thermal Resistance	θ_{JA}		45		$^{\circ}\text{C/W}$	Thermocouple located at center of package underside

¹ The device is considered a 2-terminal device; Pin 1 through Pin 8 are shorted together, and Pin 9 through Pin 16 are shorted together.

² Input capacitance is from any input data pin to ground.

SPECIFICATIONS

REGULATORY INFORMATION

The ADuM3160 certification approvals are listed in [Table 3](#). For more information about the recommended maximum working voltages for specific cross-insulation waveforms and insulation levels, see [Table 8](#) and the [Insulation Lifetime](#) section.

Table 3.

UL	CSA	VDE	CQC
UL 1577 ¹ Single protection, 2500 V _{RMS}	IEC / EN / CSA 62368-1 Basic Insulation, 600 V _{RMS} Reinforced insulation, 300 V _{RMS} IEC/CSA 61010-1 Basic Insulation, 600 V _{RMS} Reinforced insulation, 300 V _{RMS}	DIN EN IEC 60747-17 (VDE 0884-17) ² Reinforced insulation, 645 V _{PEAK}	CQC GB4943.1 Basic Insulation, 600 V _{RMS} Reinforced insulation, 300 V _{RMS}
File E214100	File No. 205078	Certificate No. 40011599	Certificate No. CQC15001129426

¹ In accordance with UL 1577, each ADuM3160 is proof tested by applying an insulation test voltage ≥ 3000 V_{RMS} for 1 sec (current leakage detection limit = 10 μ A).

² In accordance with DIN EN IEC 60747-17 (VDE 0884-17), each ADuM3160 is proof tested by applying an insulation test voltage ≥ 1208 V_{PEAK} for 1 sec (partial discharge detection limit = 5 pC). The asterisk (*) marking branded on the component indicates DIN EN IEC 60747-17 (VDE 0884-17) approval.

INSULATION AND SAFETY-RELATED SPECIFICATIONS

Table 4.

Parameter	Symbol	Value	Unit	Test Conditions/Comments
Rated Dielectric Insulation Voltage		2500	V _{RMS}	1 minute duration
Minimum External Air Gap (Clearance) ^{1, 2}	L(I01)	7.8	mm	Measured from input terminals to output terminals, shortest distance through air
Minimum External Tracking (Creepage) ¹	L(I02)	7.8	mm	Measured from input terminals to output terminals, shortest distance path along body
Minimum Internal Gap (Internal Clearance)		18	μ m	Insulation distance through insulation
Tracking Resistance (Comparative Tracking Index) ³	CTI	>400	V	DIN IEC 112/VDE 0303, Part 1
Material Group		II		Material Group (DIN VDE 0110, 1/89, Table 1)

¹ In accordance with IEC 62368-1 / IEC 60601-1 guidelines for the measurement of creepage and clearance distances for a pollution degree of 2 and altitudes ≤ 2000 m.

² Consideration must be given to pad layout to ensure the minimum required distance for clearance is maintained.

³ CTI rating for the ADuM3160 is >400 V and a Material Group II isolation group.

SPECIFICATIONS

DIN EN IEC 60747-17 (VDE 0884-17) INSULATION CHARACTERISTICS

This isolator is suitable for reinforced electrical isolation only within the safety limit data. Maintenance of the safety data is ensured by protective circuits. The asterisk (*) marking branded on the component denotes DIN EN IEC 60747-17 (VDE 0884-17) approval.

Table 5.

Description	Test Conditions/Comments	Symbol	Characteristic	Unit
Installation Classification per DIN VDE 0110			I to IV	
For Rated Mains Voltage $\leq 150 V_{RMS}$			I to III	
For Rated Mains Voltage $\leq 300 V_{RMS}$			I to II	
For Rated Mains Voltage $\leq 400 V_{RMS}$			40/105/21	
Climatic Classification			2	
Pollution Degree per DIN VDE 0110, Table 1				
Maximum Repetitive Isolation Voltage		V_{IORM}	645	V_{PEAK}
Maximum Working Isolation Voltage		V_{IOWM}	456	V_{RMS}
Input-to-Output Test Voltage, Method B1	$V_{IORM} \times 1.875 = V_{PR}$, 100% production test, $t_m = 1$ sec, partial discharge < 5 pC	V_{PR}	1209	V_{PEAK}
Input-to-Output Test Voltage, Method A		V_{PR}		
After Environmental Tests Subgroup 1	$V_{IORM} \times 1.6 = V_{PR}$, $t_m = 60$ sec, partial discharge < 5 pC		1032	V_{PEAK}
After Input and/or Safety Tests Subgroup 2 and Subgroup 3	$V_{IORM} \times 1.2 = V_{PR}$, $t_m = 60$ sec, partial discharge < 5 pC		774	V_{PEAK}
Maximum Transient Isolation Voltage	$V_{TEST} = 1.2 \times V_{IOTM}$, $t = 1$ sec (100% production)	V_{IOTM}	6000	V_{PEAK}
Maximum Impulse Voltage	Surge Voltage in Air, Waveform per IEC 61000-4-5	V_{IMP}	6000	V_{PEAK}
Maximum Surge Isolation Voltage	$V_{TEST} \geq 1.3 \times V_{IMP}$ (sample test), Tested in Oil, Waveform per IEC 61000-4-5	V_{IOSM}	10,000	V_{PEAK}
Safety-Limiting Values	Maximum value allowed in the event of a failure (see Figure 2)			
Case Temperature		T_S	150	$^{\circ}C$
Side 1 + Side 2 Current		I_{S1}	550	mA
Insulation Resistance at T_S	$V_{IO} = 500$ V	R_S	$>10^9$	Ω

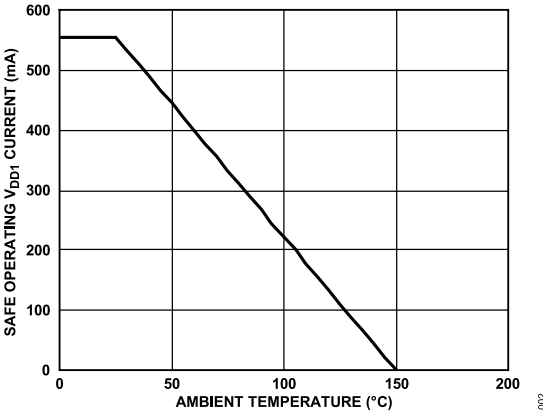


Figure 2. Thermal Derating Curve, Dependence of Safety-Limiting Values on Case Temperature, per DIN EN IEC 60747-17 (VDE 0884-17)

SPECIFICATIONS

RECOMMENDED OPERATING CONDITIONS

Table 6.

Parameter	Symbol	Min	Max	Unit
Operating Temperature	T_A	-40	+105	°C
Supply Voltages ¹	V_{BUS1} , V_{BUS2}	3.0	5.5	V
Input Signal Rise and Fall Times			1.0	ms

¹ All voltages are relative to their respective grounds. See the [DC Correctness and Magnetic Field Immunity](#) section for information about immunity to external magnetic fields.

ABSOLUTE MAXIMUM RATINGS

Ambient temperature = 25°C, unless otherwise noted.

Table 7.

Parameter	Rating
Storage Temperature (T_{ST})	-40°C to +150°C
Ambient Operating Temperature (T_A)	-40°C to +105°C
Supply Voltages (V_{BUS1} , V_{BUS2} , V_{DD1} , V_{DD2}) ^{1,2}	-0.5 V to +6.5 V
Input Voltage (V_{UD+} , V_{UD-} , V_{SPU}) ¹	-0.5 V to $V_{DD1} + 0.5$ V
Output Voltage (V_{DD-} , V_{DD+} , V_{SPD} , V_{PIN}) ¹	-0.5 V to $V_{DD2} + 0.5$ V
Average Output Current per Pin ³	
Side 1 (I_{O1})	-10 mA to +10 mA
Side 2 (I_{O2})	-10 mA to +10 mA
Common-Mode Transients ⁴	-100 kV/μs to +100 kV/μs

¹ All voltages are relative to their respective grounds.

² V_{DD1} , V_{DD2} , V_{BUS1} , and V_{BUS2} refer to the supply voltages on the input and output sides of a given channel, respectively.

³ See Figure 2 for maximum rated current values for various temperatures.

⁴ Refers to common-mode transients across the insulation barrier. Common-mode transients exceeding the Absolute Maximum Ratings may cause latch-up or permanent damage.

Stresses at or above those listed under Absolute Maximum Ratings may cause permanent damage to the product. This is a stress rating only; functional operation of the product at these or any other conditions above those indicated in the operational section of this specification is not implied. Operation beyond the maximum operating conditions for extended periods may affect product reliability.

MAXIMUM CONTINUOUS WORKING VOLTAGE

Table 8. Maximum Continuous Working Voltage¹

Parameter	Max	Unit	Constraint
AC Voltage			
Bipolar Waveform	645	V_{PEAK}	Reinforced insulation rating per IEC 60747-17 (VDE 0884-17)

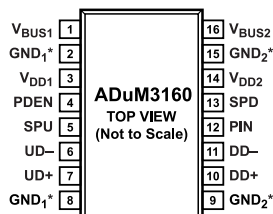
¹ Refers to continuous voltage magnitude imposed across the isolation barrier. See the [Insulation Lifetime](#) section for more information.

ESD CAUTION



ESD (electrostatic discharge) sensitive device. Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

PIN CONFIGURATION AND FUNCTION DESCRIPTIONS



*PIN 2 AND PIN 8 ARE INTERNALLY CONNECTED TO EACH OTHER, AND IT IS RECOMMENDED THAT BOTH PINS BE CONNECTED TO A COMMON GROUND. PIN 9 AND PIN 15 ARE INTERNALLY CONNECTED TO EACH OTHER, AND IT IS RECOMMENDED THAT BOTH PINS BE CONNECTED TO A COMMON GROUND.

003

Figure 3. Pin Configuration

Table 9. Pin Function Descriptions

Pin No.	Mnemonic	Direction	Description
1	V _{BUS1}	Power	Input Power Supply for Side 1. When the isolator is powered by the USB bus voltage (4.5 V to 5.5 V), connect the V _{BUS1} pin to the USB power bus. When the isolator is powered from a 3.3 V power supply, connect V _{BUS1} to V _{DD1} and to the external 3.3 V power supply. A bypass capacitor to GND ₁ is required.
2	GND ₁	Return	Ground 1. Ground reference for Isolator Side 1. Pin 2 and Pin 8 are internally connected to each other, and it is recommended that both pins be connected to a common ground.
3	V _{DD1}	Power	Input Power Supply for Side 1. When the isolator is powered by the USB bus voltage (4.5 V to 5.5 V), the V _{DD1} pin should be used for a bypass capacitor to GND ₁ . Signal lines that may require pull-up, such as PDEN and SPU, should be tied to this pin. When the isolator is powered from a 3.3 V power supply, connect V _{BUS1} to V _{DD1} and to the external 3.3 V power supply. A bypass capacitor to GND ₁ is required.
4	PDEN	Input	Pull-Down Enable. This pin is read when exiting reset. This pin must be connected to V _{DD1} for standard operation. If this pin is connected to GND ₁ when exiting reset, the downstream pull-down resistors are disconnected, allowing buffer impedance measurements.
5	SPU	Input	Speed Select, Upstream Buffer. Active high logic input. When SPU is tied high, the full speed slew rate, timing, and logic conventions are selected; when SPU is tied low, the low speed slew rate, timing, and logic conventions are selected. This input must be set high via connection to V _{DD1} or set low via connection to GND ₁ and must match Pin 13 (both pins tied high or both pins tied low).
6	UD-	Input/Output	Upstream D-.
7	UD+	Input/Output	Upstream D+.
8	GND ₁	Return	Ground 1. Ground reference for Isolator Side 1. Pin 2 and Pin 8 are internally connected to each other, and it is recommended that both pins be connected to a common ground.
9	GND ₂	Return	Ground 2. Ground reference for Isolator Side 2. Pin 9 and Pin 15 are internally connected to each other, and it is recommended that both pins be connected to a common ground.
10	DD+	Input/Output	Downstream D+.
11	DD-	Input/Output	Downstream D-.
12	PIN	Input	Upstream Pull-Up Enable. PIN controls the power connection to the pull-up for the upstream port. It can be tied to V _{DD2} for operation on power-up, or it can be tied to an external control signal for an application that requires delayed enumeration.
13	SPD	Input	Speed Select, Downstream Buffer. Active high logic input. When SPD is tied high, the full speed slew rate, timing, and logic conventions are selected; when SPD is tied low, the low speed slew rate, timing, and logic conventions are selected. This input must be set high via connection to V _{DD2} or set low via connection to GND ₂ and must match Pin 5 (both pins tied high or both pins tied low).
14	V _{DD2}	Power	Input Power Supply for Side 2. When the isolator is powered by the USB bus voltage (4.5 V to 5.5 V), the V _{DD2} pin should be used for a bypass capacitor to GND ₂ . Signal lines that may require pull-up, such as SPD, can be tied to this pin. When the isolator is powered from a 3.3 V power supply, connect V _{BUS2} to V _{DD2} and to the external 3.3 V power supply. A bypass capacitor to GND ₂ is required.
15	GND ₂	Return	Ground 2. Ground reference for Isolator Side 2. Pin 9 and Pin 15 are internally connected to each other, and it is recommended that both pins be connected to a common ground.
16	V _{BUS2}	Power	Input Power Supply for Side 2. When the isolator is powered by the USB bus voltage (4.5 V to 5.5 V), connect the V _{BUS2} pin to the USB power bus. When the isolator is powered from a 3.3 V power supply, connect V _{BUS2} to V _{DD2} and to the external 3.3 V power supply. A bypass capacitor to GND ₂ is required.

PIN CONFIGURATION AND FUNCTION DESCRIPTIONS

Table 10. Truth Table, Control Signals, and Power (Positive Logic)

V_{SPU} Input ¹	V_{UD+}, V_{UD-} State ¹	V_{BUS1}, V_{DD1} State	V_{BUS2}, V_{DD2} State	V_{DD+}, V_{DD-} State ¹	V_{PIN} Input ¹	V_{SPD} Input ¹	Description
High	Active	Powered	Powered	Active	High	High	Input and output logic set for full speed logic convention and timing.
Low	Active	Powered	Powered	Active	High	Low	Input and output logic set for low speed logic convention and timing.
Low	Active	Powered	Powered	Active	High	High	Not allowed. V_{SPU} and V_{SPD} must be set to the same value. The USB host detects a communication error.
High	Active	Powered	Powered	Active	High	Low	Not allowed. V_{SPU} and V_{SPD} must be set to the same value. The USB host detects a communication error.
X	Z	Powered	Powered	Z	Low	X	Upstream Side 1 presents a disconnected state to the USB cable.
X	X	Unpowered	Powered	Z	X	X	When power is not present on V_{DD1} , the down-stream data output drivers revert to the high-Z state within 32 bit times. The downstream side initializes in the high-Z state.
X	Z	Powered	Unpowered	X	X	X	When power is not present on V_{DD2} , the upstream side disconnects the pull-up and disables the upstream drivers within 32 bit times.

¹ X is don't care; Z is the high impedance output state.

APPLICATIONS INFORMATION

FUNCTIONAL DESCRIPTION

USB isolation in the D+/D- lines is challenging for several reasons. First, access to the output enable signals is normally required to control the transceiver. Some level of intelligence must be built into the isolator to interpret the data stream and determine when to enable and disable its upstream and down-stream output buffers. Second, the signal must be faithfully reconstructed on the output side of the coupler while retaining precise timing and not passing transient states such as invalid SE0 and SE1 states. In addition, the part must meet the low power requirements of the suspend mode.

The iCoupler technology is based on edge detection and, therefore, lends itself well to the USB application. The flow of data through the device is accomplished by monitoring the inputs for activity and setting the direction for data transfer based on a transition from the idle state. After data directionality is established, data is transferred until either an end of packet (EOP) or a sufficiently long idle state is encountered. At this point, the coupler disables its output buffers and monitors its inputs for the next activity.

During the data transfers, the input side of the coupler holds its output buffers disabled. The output side enables its output buffers and disables edge detection from the input buffers. This allows the data to flow in one direction without wrapping back through the coupler, causing the iCoupler to latch. Timing is based on the differential input signal transition. Logic is included to eliminate any artifacts due to different input thresholds of the differential and single-ended buffers. The input state is transferred across the isolation barrier as one of three valid states, J, K, or SE0. The signal is reconstructed at the output side with a fixed time delay from the input side differential input.

The iCoupler does not have a special suspend mode, nor does it need one because its power supply current is below the suspend current limit of 2.5 mA when the USB bus is idle.

The ADuM3160 is designed to interface with an upstream-facing low/full speed USB port by isolating the D+/D- lines. An upstream-facing port supports only one speed of operation; therefore, the speed-related parameters, J/K logic levels, and D+/D- slew rate are set to match the speed of the upstream-facing peripheral port (see Table 10).

A control line on the downstream side of the ADuM3160 activates the idle state pull-up resistor. This allows the downstream port to control when the upstream port attaches to the USB bus. The PIN input can be tied to a 3.3 V control logic signal or to the V_{DD2} rail depending on whether enumeration must be controlled or will occur when power is first applied.

PRODUCT USAGE

The ADuM3160 is designed to be integrated into a USB peripheral with an upstream-facing USB port, as shown in Figure 4. The key design points are as follows:

- ▶ The USB host provides power for the upstream side of the ADuM3160 through the cable.
- ▶ The peripheral supply provides power to the downstream side of the ADuM3160.
- ▶ The DD+/DD- lines of the isolator interface with the peripheral controller, and the UD+/UD- lines of the isolator connect to the cable or host.
- ▶ Peripheral devices have a fixed data rate that is set at design time. The ADuM3160 has configuration pins, SPU and SPD, that are set by the user to match this speed on the upstream and downstream sides of the coupler.
- ▶ USB enumeration begins when either the D+ or D- line is pulled high at the peripheral end of the USB cable. Control of the timing of this event is provided by the PIN input on the downstream side of the coupler.
- ▶ Pull-up and pull-down resistors are implemented inside the coupler. Only external series resistors and bypass capacitors are required for operation.

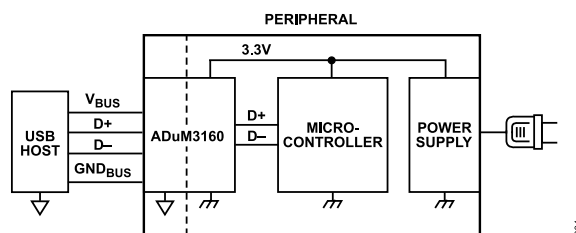


Figure 4. Typical ADuM3160 Application

Other than the delayed application of pull-up resistors, the ADuM3160 is transparent to USB traffic, and no modifications to the peripheral design are required to provide isolation. The isolator adds propagation delay to the signals comparable to a hub and cable. Isolated peripherals must be treated as if there were a built-in hub when determining the maximum number of hubs in a data chain.

Hubs can be isolated like any other peripheral. Isolated hubs can be created by placing an ADuM3160 on the upstream port of a hub chip. This configuration can be made compliant if counted as two hub delays. The hub chip allows the ADuM3160 to operate at full speed yet maintain compatibility with low speed devices.

APPLICATIONS INFORMATION

COMPATIBILITY OF UPSTREAM APPLICATIONS

The ADuM3160 is designed specifically for isolating a USB peripheral. However, the chip has two USB interfaces that meet the electrical requirements for driving USB cables. This opens the possibility of implementing isolation in downstream USB ports such as isolated cables, which have generic connections to both upstream and downstream devices, as well as isolating host ports.

In a fully compliant application, a downstream-facing port must be able to detect whether a peripheral is low speed or full speed based on the application of the upstream pull-up. The buffers and logic conventions must adjust to match the requested speed. Because the ADuM3160 sets its speed by hardwiring pins, the part cannot adjust to different peripherals on the fly.

The practical result of using the ADuM3160 in a host port is that the port works at a single speed. This behavior is acceptable in embedded host applications; however, this type of interface is not fully compliant as a general-purpose USB port.

Isolated cable applications have a similar issue. The cable operates at the preset speed only; therefore, treat cable assemblies as custom applications, not general-purpose isolated cables.

POWER SUPPLY OPTIONS

In most USB transceivers, 3.3 V is derived from the 5 V USB bus through an LDO regulator. The ADuM3160 includes internal LDO regulators on both the upstream and downstream sides. The output of the LDO regulators is available on the V_{DD1} and V_{DD2} pins. In some cases, especially on the peripheral side of the isolation, there may not be a 5 V power supply available. The ADuM3160 has the ability to bypass the regulator and run on a 3.3 V supply directly.

Two power pins are present on each side, V_{BUSx} and V_{DDx} . If 5 V is supplied to V_{BUSx} , an internal regulator creates 3.3 V to power the $x\text{D}+$ and $x\text{D}-$ drivers. V_{DDx} provides external access to the 3.3 V supply to allow external bypass as well as bias for external pull-ups. If only 3.3 V is available, it can be supplied to both V_{BUSx} and V_{DDx} . This disables the regulator and powers the coupler directly from the 3.3 V supply.

Figure 5 shows how to configure a typical application when the upstream side of the coupler receives power directly from the USB bus and the downstream side receives 3.3 V from the peripheral power supply. The downstream side can run from a 5 V V_{BUS2} power supply as well. It can be connected in the same manner as V_{BUS1} , as shown in Figure 5, if needed.

PRINTED CIRCUIT BOARD LAYOUT

The ADuM3160 digital isolator requires no external interface circuitry for the logic interfaces. For full speed operation, the $\text{D}+$ and $\text{D}-$ lines on each side of the device require a $24\ \Omega \pm 1\%$ series termination resistor. These resistors are not required for low speed applications. Power supply bypassing is required at the input

and output supply pins (see Figure 5). Install bypass capacitors between V_{BUSx} and V_{DDx} on each side of the chip. The capacitors should have a minimum value of $0.1\ \mu\text{F}$ and low ESR. The total lead length between both ends of the capacitor and the power supply pin should not exceed 10 mm.

Bypassing between Pin 2 and Pin 8 and between Pin 9 and Pin 15 should also be considered unless the ground pair on each package side is connected close to the package. All logic level signals are 3.3 V and should be referenced to the local V_{DDx} pin or 3.3 V logic signals from an external source.

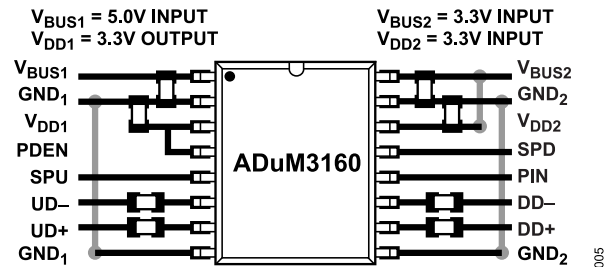


Figure 5. Suggested PCB Layout Example

In applications that involve high common-mode transients, care should be taken to minimize board coupling across the isolation barrier. Furthermore, design the board layout such that any coupling that does occur affects all pins equally on a given component side. Failure to ensure this can cause voltage differentials between pins that exceed the absolute maximum ratings of the device, thereby leading to latch-up or permanent damage.

DC CORRECTNESS AND MAGNETIC FIELD IMMUNITY

Positive and negative logic transitions at the isolator input cause narrow ($\sim 1\ \text{ns}$) pulses to be sent to the decoder via the transformer. The decoder is bistable and is, therefore, either set or reset by the pulses, indicating input logic transitions.

The limitation on the magnetic field immunity of the ADuM3160 is set by the condition in which induced voltage in the receiving coil of the transformer is sufficiently large to either falsely set or reset the decoder. The following analysis defines the conditions under which this may occur. The 3 V operating condition of the ADuM3160 is examined because it represents the most susceptible mode of operation.

The pulses at the transformer output have an amplitude greater than 1.0 V. The decoder has a sensing threshold at approximately 0.5 V, thus establishing a 0.5 V margin in which induced voltages can be tolerated. The voltage induced across the receiving coil is given by

$$V = (-d\beta/dt) \sum \pi r_n^2; n = 1, 2, \dots, N \quad (1)$$

where:

β is the magnetic flux density (gauss).

r_n is the radius of the n^{th} turn in the receiving coil (cm).

APPLICATIONS INFORMATION

N is the number of turns in the receiving coil.

Given the geometry of the receiving coil in the ADuM3160 and an imposed requirement that the induced voltage be, at most, 50% of the 0.5 V margin at the decoder, a maximum allowable magnetic field is calculated as shown in Figure 6.

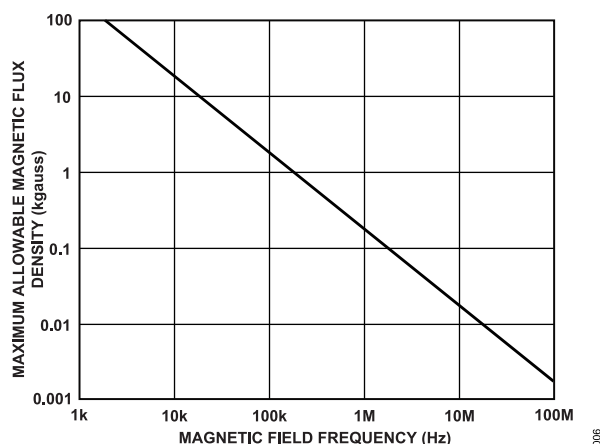


Figure 6. Maximum Allowable External Magnetic Flux Density

For example, at a magnetic field frequency of 1 MHz, the maximum allowable magnetic field of 0.2 kgauss induces a voltage of 0.25 V at the receiving coil. This voltage is approximately 50% of the sensing threshold and does not cause a faulty output transition. Similarly, if such an event occurs during a transmitted pulse (and is of the worst-case polarity), it reduces the received pulse from >1.0 V to 0.75 V—still well above the 0.5 V sensing threshold of the decoder.

The preceding magnetic flux density values correspond to specific current magnitudes at given distances from the ADuM3160 transformers. Figure 7 expresses these allowable current magnitudes as a function of frequency for selected distances. As shown in Figure 7, the ADuM3160 is extremely immune and can be affected only by extremely large currents operated at high frequency very close to the component. For the 1 MHz example noted, a 0.5 kA current must be placed 5 mm away from the ADuM3160 to affect the operation of the device.

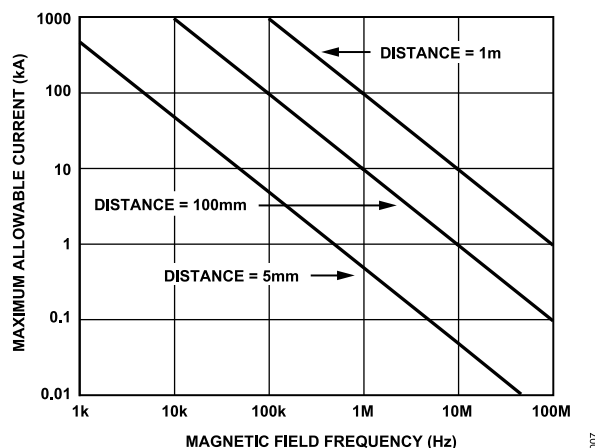


Figure 7. Maximum Allowable Current for Various Current-to-ADuM3160 Spacings

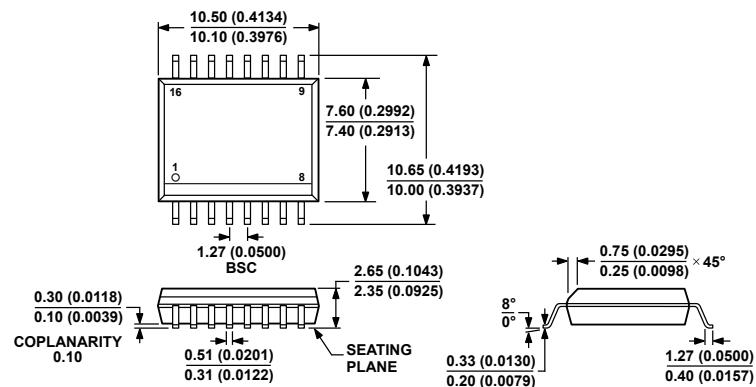
Note that at combinations of strong magnetic field and high frequency, any loops formed by PCB traces may induce error voltages sufficiently large to trigger the thresholds of succeeding circuitry. Care should be taken in the layout of such traces to avoid this possibility.

INSULATION LIFETIME

All insulation structures eventually break down when subjected to voltage stress over a sufficiently long period. The rate of insulation degradation is dependent on the characteristics of the voltage waveform applied across the insulation. In addition to the testing performed by the regulatory agencies, Analog Devices carries out an extensive set of evaluations to determine the lifetime of the insulation structure within the ADuM3160.

Analog Devices performs accelerated life testing using voltage levels higher than the rated continuous working voltage. Acceleration factors for several operating conditions are determined. These factors allow calculation of the time to failure at the actual working voltage. The values shown in Table 8 summarize the maximum continuous working voltages as per IEC 60747-17. Operation at these high working voltages can lead to shortened insulation life in some cases.

OUTLINE DIMENSIONS



COMPLIANT TO JEDEC STANDARDS MS-013-AA
CONTROLLING DIMENSIONS ARE IN MILLIMETERS; INCH DIMENSIONS
(IN PARENTHESES) ARE ROUNDED-OFF MILLIMETER EQUIVALENTS FOR
REFERENCE ONLY AND ARE NOT APPROPRIATE FOR USE IN DESIGN.

03-27-2007-B

Figure 8. 16-Lead Standard Small Outline Package [SOIC_W]

Wide Body

(RW-16)

Dimensions shown in millimeters and (inches)

ORDERING GUIDE

Model ^{1,2}	Temperature Range	Package Description	Packing Quantity	Package Option
ADuM3160BRWZ	-40°C to +105°C	16-Lead SOIC_W	Tube, 47	RW-16
ADuM3160BRWZ-RL	-40°C to +105°C	16-Lead SOIC_W	Reel, 1000	RW-16
ADuM3160WBRWZ	-40°C to +105°C	16-Lead SOIC_W	Tube, 47	RW-16
ADuM3160WBRWZ-RL	-40°C to +105°C	16-Lead SOIC_W	Reel, 1000	RW-16

¹ Z = RoHS Compliant Part.

² W = Qualified for Automotive Applications.

NUMBER OF INPUTS, MAXIMUM FULL SPEED DATA RATE, MAXIMUM FULL SPEED PROPAGATION DELAY, AND MAXIMUM FULL SPEED JITTER OPTIONS

Model ^{1,2}	Number of Inputs, V _{DD1} Side	Number of Inputs, V _{DD2} Side	Maximum Full Speed Data Rate (Mbps)	Maximum Full Speed Propagation Delay, 5 V (ns)	Maximum Full Speed Jitter (ns)
ADuM3160BRWZ	2	2	12	70	3
ADuM3160BRWZ-RL	2	2	12	70	3
ADuM3160WBRWZ	2	2	12	70	3
ADuM3160WBRWZ-RL	2	2	12	70	3

¹ Z = RoHS Compliant Part.

² W = Qualified for Automotive Applications.

EVALUATION BOARDS

Model ¹	Description
EVAL-ADUM4160EBZ	Evaluation Board

¹ Z = RoHS Compliant Part.

OUTLINE DIMENSIONS

AUTOMOTIVE PRODUCTS

The ADuM3160W model is available with controlled manufacturing to support the quality and reliability requirements of automotive applications. Note that these automotive models may have specifications that differ from the commercial models; therefore, designers should review the [Specifications](#) section of this data sheet carefully. Only the automotive grade products shown are available for use in automotive applications. Contact your local Analog Devices account representative for specific product ordering information and to obtain the specific Automotive Reliability reports for these models.